

TOMÁS S. R. SILVA

Curriculum Vitæ

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Introduction. *I am a PhD candidate in Mathematics at the Institute of Mathematics, Statistics and Scientific Computing (IMECC), University of Campinas (UNICAMP), Brazil. My research applies machine learning and artificial intelligence techniques to problems in complex differential and algebraic geometry.*

EDUCATION

PhD, Mathematics Aug 2022 – Feb 2027 (expected)
 University of Campinas (UNICAMP) Campinas, São Paulo, Brazil
 Supervisor: Prof. Henrique N. Sá Earp

BSc, Computer Engineering Mar 2017 – Dec 2021
 University of Campinas (UNICAMP) Campinas, São Paulo, Brazil
 Supervisor: Prof. Ricardo Dahab

RESEARCH EXPERIENCE

Visiting Research Student Jan 2024 – Jul 2024
 City, University of London London, UK
 Supervisor: Prof. Yang-Hui He (London Institute for Mathematical Sciences & Merton College, Oxford)

PhD Researcher Aug 2022 – present
 Laboratory of Gauge Theory and Algebraic Geometry (GTAG), UNICAMP

Junior Researcher May 2018 – Dec 2021
 Security and Cryptography Laboratory (LASCA), UNICAMP

PUBLICATIONS

• Peer-Reviewed Journal Articles

The exterior derivative and the mean value equality in $\mathbb{R}^n$ 2026
 D. Fadel, H. N. Sá Earp & [T. Silva](#)
[arXiv:2510.00999](#) — to appear in *São Paulo Journal of Mathematical Sciences*.

Neural and numerical methods for G_2 -structures on contact Calabi–Yau 7-manifolds 2026
 E. Heyes, E. Hirst, H. N. Sá Earp & [T. Silva](#)
Physics Letters B, Vol. 878, 2026. $\diamond$ [arXiv:2602.12438](#)

Metaheuristic Generation of Brane Tilings 2025
 Y.-H. He, V. Jejjala & [T. Silva](#)
Physics Letters B, Vol. 862, 2025. $\diamond$ [arXiv:2412.19313](#)

Efficient isochronous fixed-weight sampling with applications to NTRU 2024
 D. L. Gazzoni Filho, [T. Silva](#) & J. López
IACR Communications in Cryptology, Vol. 1, No. 2, 2024.

Machine-learning Sasakian and G_2 topology on contact Calabi–Yau 7-manifolds 2024
 D. Aggarwal, Y.-H. He, E. Heyes, E. Hirst, H. N. Sá Earp & [T. Silva](#)
Physics Letters B, Vol. 850, 2024. $\diamond$ [arXiv:2310.03064](#)

• Peer-Reviewed Conference Proceedings

Machine learning topology of Calabi–Yau links 2025
[T. Silva](#) & H. N. Sá Earp
Proceeding Series of the Brazilian Society of Computational and Applied Mathematics, Vol. 11, No. 1, 2025.

• Preprints

Minimising Willmore Energy via Neural Flow 2026
E. Hirst, H. N. Sá Earp & T. Silva
[arXiv:2604.04321](https://arxiv.org/abs/2604.04321)

A semicontinuous relaxation of Saito's criterion and freeness as angular minimization 2026
T. Silva
[arXiv:2604.02995](https://arxiv.org/abs/2604.02995)

HONOURS & AWARDS

Academic Distinction 2022
Awarded by the Institute of Computing, UNICAMP, for academic excellence during the undergraduate programme.

Best Undergraduate Research Project 2021
Awarded by the Institute of Computing, UNICAMP, for the work “*Characterizing Lattices for Cryptography*”, recognised as a significant contribution to scientific and technological research in Computing that year.

INVITED TALKS

Mathematics as a game: exploring freeness of line arrangements with machine learning 2026
Mathematics Department Seminar, Institute of Mathematical and Computer Sciences (ICMC), University of São Paulo (USP).

A data-driven view of exterior calculus 2026
DANGER: Data, Numbers, and Geometry — Banff International Research Station (BIRS).

From Theory to Computation: Unravelling Hyperplane Arrangements 2025
Geometry, Algebra and Combinatorics Seminar, UNICAMP.

Machine Learning Ricci-Flat Metrics 2025
São Paulo Meeting on Geometry, Institute of Mathematics and Statistics (IME), University of São Paulo (USP); Geometry Seminar, Institute of Mathematical and Computer Sciences (ICMC), University of São Paulo (USP); AI+Math Seminar, Brazilian Institute of Data Science (BIOS).

Machine Learning Riemannian Metrics with Special Holonomy 2025
Machine Learning and Mathematics programme, Korea Institute for Advanced Study (KIAS).

From Theory to Computation: Unravelling Hyperplane Arrangements 2024
Seminar on Commutative Algebra and Algebraic Geometry, Federal University of Pernambuco (UFPE).

Learning Geometry and Topology of Algebraic Varieties 2023
AI+Math Seminar, Brazilian Institute of Data Science (BIOS).

GRANTS & FELLOWSHIPS

- Proposal Title: Geometry, Topology and Data Science
Funding Agency: São Paulo Research Foundation (FAPESP), grant no. 2022/09891-4, within the FAPESP-ANR Thematic Project *BRIDGES: France-Brazil Interactions in Gauge Theory, Extremal Structures and Stability*, grant no. 2021/04065-6
Role: Doctoral fellow | Principal Investigator: Prof. Henrique N. Sá Earp
Coverage Period: Sep 2022 – present
- Proposal Title: Post-Quantum Cryptography
Funding Agency: Samsung R&D Institute Brazil
Role: Undergraduate research grantee | Principal Investigators: Prof. Ricardo Dahab & Prof. Julio López
Coverage Period: Nov 2020 – Nov 2021
- Proposal Title: Characterizing Lattices and Codes for Cryptography
Funding Agency: São Paulo Research Foundation (FAPESP), grant no. 2019/19296-3
Role: Undergraduate research grantee | Principal Investigator: Prof. Ricardo Dahab
Coverage Period: Nov 2019 – Oct 2020
- Proposal Title: Characterizing Lattices for Cryptography
Funding Agencies: São Paulo Research Foundation (FAPESP), grant no. 2018/08130-4; and National Council for Scientific and Technological Development (CNPq), grant no. 124431/2019-4
Role: Undergraduate research grantee | Principal Investigator: Prof. Ricardo Dahab
Coverage Period: May 2018 – Apr 2019 (FAPESP) & Aug 2019 – Nov 2019 (CNPq)

TEACHING & ADVISING

Courses — Teaching Assistant (UNICAMP)

Linear Algebra Aug 2025 – Dec 2025 & Feb 2026 – Jul 2026
Vector Calculus (Calculus II) Feb 2025 – Jul 2025 & Feb 2026 – Jul 2026
Analytic Geometry Aug 2022 – Dec 2022 & Aug 2023 – Dec 2023

Undergraduate Student Co-Supervision

Bianca Bosnardo April 2026 – March 2027
Undergrad research project: “An introduction to hyperplane arrangements”.

SERVICE & PROFESSIONAL DEVELOPMENT

• Service to the Discipline and Profession

Founder and Moderator — MathIA AmLat 2026–present

Founder and coordinator of a Latin American community dedicated to discussion and collaboration on applications of artificial intelligence in mathematical research. math-ia-amlat.zulipchat.com

Co-organiser — “AI meets Geometry” Workshop 2026

XXII Escola de Geometria Diferencial (EGD). egd.ufpi.edu.br

• Service to the Community and Public

Volunteer Mathematics Tutor — “Cursinho Popular Zilda Arns” Jan 2020 – Jan 2021

A non-profit preparatory school supporting underprivileged students in preparing for Brazilian university entrance exams.

• Professional Development — Invited Programme Participation

Actions and Intuition: Algebra via AI,
Hausdorff Research Institute for Mathematics (HIM) Sep 2026
AI and Mathematics,
Korea Institute for Advanced Study (KIAS) Jul–Aug 2026
Mathematics and Machine Learning Program,
Harvard CMSA Sep–Nov 2024

• Professional Development — Workshops & Conferences Attended

Inaugural Workshop of the Brazilian Center for Geometry, UNICAMP 2026
Vector Bundles and their Applications in Algebraic Geometry and
Related Topics (Bandoleros), UNICAMP 2025
XLIII National Congress of Applied and Computational Mathematics,
Federal University of Pernambuco 2024
London Geometry and Machine Learning (LOGML), Imperial College London 2024
Bridges Meeting on Special Geometries and Stability, Institut d’Études Scientifiques, Cargèse 2024
Hybrid Conference on AI-Mathematics, Rio de Janeiro State University 2024
MATH-AmSud School on Geometry: Group Actions, Symmetries, Moduli and Beyond,
University of Chile 2023
EnCoRi — Encounter on Codes, Lattices and Information Theory, UNICAMP 2023
XV ALGA — Commutative Algebra and Algebraic Geometry, IMPA 2023
XVI Theses, Dissertations and Undergraduate Research Workshop, Computing Institute, UNICAMP 2021
XV Theses, Dissertations and Scientific Undergraduate Research, Computing Institute, UNICAMP .. 2020
I Journey of Undergraduate Research in Codes, Cryptography and Information Theory, UNICAMP .. 2019
Metric and Combinatorial Problems Related to ECC, UNICAMP 2019
XIV Theses, Dissertations and Undergraduate Research Workshop, Computing Institute, UNICAMP 2019
XXVII Congress of Undergraduate Research, UNICAMP 2019
I Latin American Week on Coding and Information, UNICAMP 2018

LANGUAGES & TECHNICAL SKILLS

Languages: Portuguese (native), English (advanced), Spanish (advanced), Hebrew (intermediate), French (intermediate).

Programming: C, C++, Python (PyTorch/JAX/TensorFlow), CUDA, Wolfram Mathematica, SageMath, Julia, SQL, Haskell, Linux/Bash, ARM Assembly.

Document Creation: $\text{\LaTeX}$, Microsoft Office Suite, Markdown.